FUZZY LOGIC & NEURAL NETWORKS · SEMINAR 2

Lung Cancer Classification

From Data Leakage Diagnosis to Patient-Level LUNA16 Models

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Why Lung Cancer Classification Matters

Leading cause of cancer death

≈ 1.8 million deaths worldwide each year (WHO).

Early detection is decisive

5-year survival jumps from ~5 % (late stage) to ~60 % when detected via low-dose CT screening.

Manual screening bottleneck

Radiologists must inspect 200–400 slices per patient — slow, tiring, and inter-reader variable.

OUR TASK

Binary nodule classification

Input: 64 × 64 CT patch (50 mm region)

Output: nodule / non-nodule

Goal: leakage-free, clinically credible classifier

Why this Project — and What we Aim to Deliver

Motivations

- Reduce radiologist workload
- Improve detection consistency
- Provide a clinically credible benchmark — not an inflated one

Aims

- Train a CNN for binary nodule classification
- Establish a leakage-free evaluation protocol
- Compare regularisation & architecture choices systematically

Objectives

- Patient-level data split (no patient overlap)
- 9-metric suite: Acc, Bal-Acc, Prec, Rec, Spec, F1, MCC, ROC-AUC, PR-AUC
- Iterative improvement: baseline → +regularisation → backbone swap

From “Too Good to be True” to a Trustworthy Result

PHASE 1

IQ-OTH/NCCD

First attempt — suspicion

- 1 097 CT slices, 3 classes
- ResNet-50 baseline → 99 %+ accuracy
- “Too good to be true” warning

PHASE 2

Investigation

Patient-level leakage

- Pixel-correlation analysis between train and test
- Found near-identical adjacent slices in both sets
- Inflated metrics confirmed — discard random split

PHASE 3

Pivot to LUNA16

Patient-level evaluation

- 888 patients, candidate-based annotations
- Patient-level split: subsets 0–6 train, 7 val, 8+9 test
- Honest, comparable metrics

End-to-End Pipeline

Raw CT Volumes
.mhd / .raw
888 patients

→

Patch Extraction
World→Voxel
50 mm · 64×64
HU [-1000, 400]

→

Augmentation
Flip · Rotate
Erasing · MixUp

→

Backbone + CBAM
ResNet-50 / DenseNet-121
Channel + Spatial Attn.

→

Classifier Head
BN · Dropout 0.4
Linear(2)

TRAINING

Focal Loss
α = 0.25 · γ = 2.0

AdamW
wd = 1e-4 · clip 1.0

OneCycleLR
max_lr = 5e-4 · cos anneal

Mixed Precision
AMP + Early Stop (val F1)

INFERENCE

5-View Test-Time Augmentation: identity · h-flip · v-flip · rot90 · rot270 → averaged softmax → final prediction

Three Versions, Identical Recipe

Backbone & Head

Input 64×64 (resized 224, 3-ch grayscale)

Backbone (ImageNet V2 pretrained)

ResNet-50 (v1, v2) · DenseNet-121 (v3)

CBAM Attention (v2, v3 only)

Channel Attention → Spatial Attention

Adaptive AvgPool 1×1

Head: BatchNorm + Dropout 0.4 + Linear(2)

The three model variants

v1 · Baseline

ResNet-50 · weighted CE · no MixUp · no TTA · 30 epochs

v2 · Improved

ResNet-50 + CBAM + Focal + MixUp + TTA · 25 epochs

v3 · Backbone Swap

DenseNet-121 + CBAM + Focal + MixUp + TTA · 25 epochs

Exactly What Changes from v1 → v2 → v3

Component	v1 · Baseline	v2 · Improved	v3 · Backbone Swap
Backbone	ResNet-50	ResNet-50	**DenseNet-121**
Parameters	23.5 M	24.0 M	**7.0 M (3× smaller)**
CBAM Attention	—	✓	✓
Loss Function	Weighted CE	**Focal Loss (α=0.25, γ=2)**	**Focal Loss (α=0.25, γ=2)**
MixUp Augmentation	—	✓	✓
Test-Time Augmentation	—	**5-view averaged**	**5-view averaged**
Epochs	30	25	25

How to read this lineage

v2 = v1 + the four improvements (CBAM attention · Focal loss · MixUp augmentation · TTA) that directly target ResNet-50's known limitations:
no explicit attention, naive treatment of class imbalance, weak regularisation, single-pass inference.
v3 = v2 with the backbone swapped to a smaller, parameter-efficient architecture.

Dataset and Patient-Level Split

TRAIN

4 972

patches · 623 patients
Subsets 0–6

VALIDATION

585

patches · 89 patients
Subset 7

TEST

1 528

patches · 176 patients
Subsets 8 + 9

Why this split (Option B)

Patient-level partitioning — no patient appears in two splits.

Test set doubled from 662 → 1 528 patches ⇒ CI on F1 narrows from ±0.012 to **±0.008**

→ smaller architectural deltas now become *statistically detectable.*

Class prevalence is consistent across splits (≈ 22 % nodule), so train-time class imbalance handling generalises to evaluation.

Test Metrics on 1 528 Held-Out Patches

Metric	v1 (single)	v2 (+TTA)	v3 (+TTA)
Accuracy	**0.9738**	0.9732	0.9692
Balanced Accuracy	**0.9577**	0.9551	0.9537
Precision	**0.9800**	0.9784	0.9791
Recall (Sensitivity)	0.9866	**0.9874**	0.9815
Specificity	**0.9288**	0.9228	0.9258
F1-Score	**0.9833**	0.9829	0.9803
MCC	**0.9233**	0.9212	0.9103
ROC-AUC	**0.9929**	0.9917	0.9896
PR-AUC	**0.9978**	0.9966	0.9960

How to read this

Red bold = best value in each row

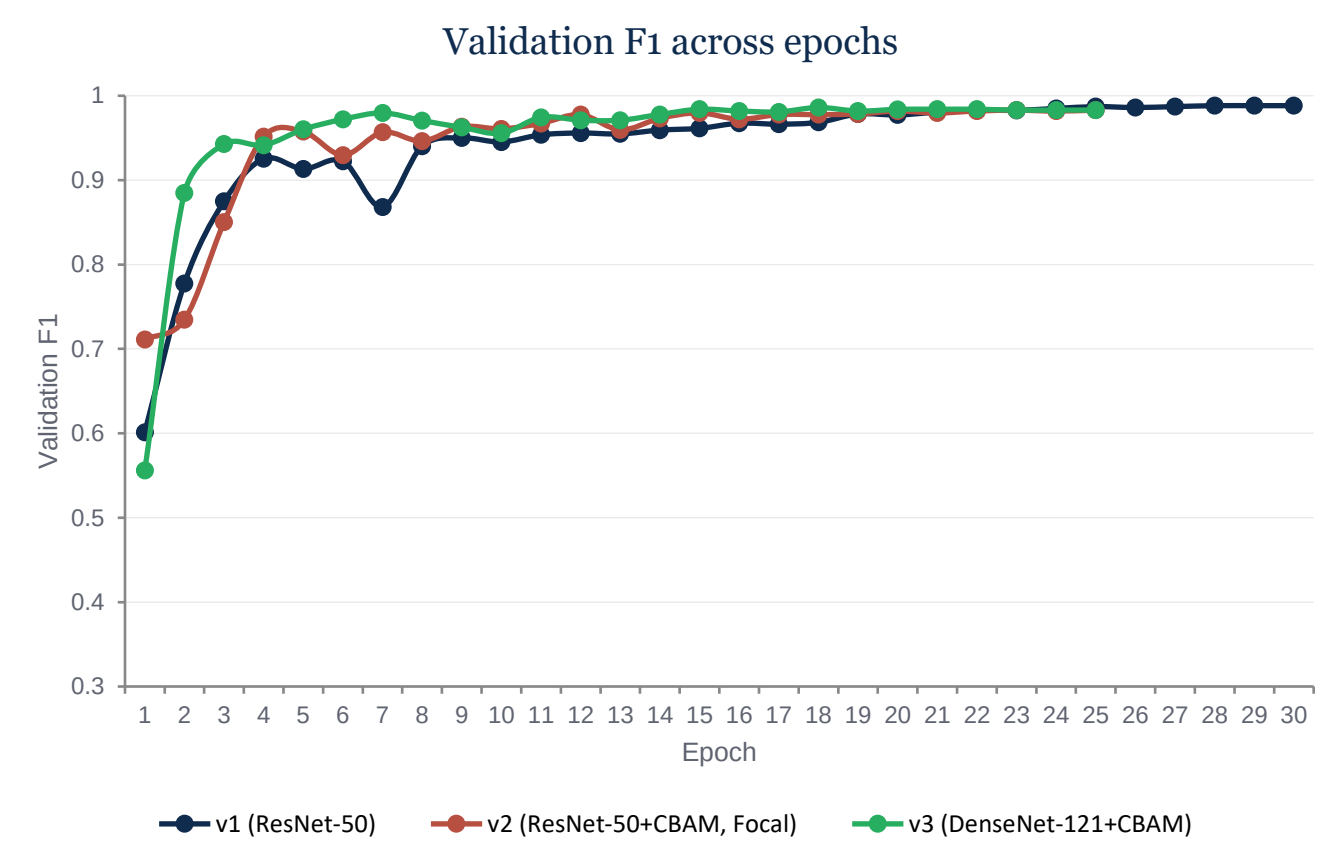
v1 wins 7 of 9 metrics

v2 wins Recall

v3 wins Precision

BUT — all deltas fit *inside the ±0.008 F1 confidence interval.*

All Three Models Reach the Same Ceiling



Confusion (v1, best F1 = 0.9833)

	Pred. non-nodule	Pred. nodule
True non-nodule	313	24
True nodule	16	1175

Over/underfitting check

v1: train 98.8% / val 98.1%, gap +0.7% ✓
v2: val ≥ train (MixUp signature) ✓
v3: val ≥ train (MixUp signature) ✓

Honest Findings — what we learned

FINDING 1

All three models hit the same ceiling.

F1 ≈ 0.98, AUC ≈ 0.99, MCC ≈ 0.92 across v1, v2, v3.

Largest pair-wise F1 delta: **0.003** — smaller than the **±0.008** confidence interval at this test size.

Statistically: *indistinguishable.*

FINDING 2

Data > architecture at this scale.

Earlier 1 407-patch run: **v2 beat v1 by +2.6 F1.**
Now (4 972 patches): **the gap vanishes.**

Regularisation tricks (CBAM, Focal, MixUp) help most when training data is scarce.

With sufficient data, plain ResNet-50 is enough.

Take-away: *more data moved the needle far more than any architectural trick.*

What Each Improvement Contributes Alone

Variant	F1	MCC	ROC-AUC	Δ F1 vs v1	Verdict
v1 (baseline)	0.9833	0.9233	0.9929	—	*reference*
v1 + CBAM	0.9795	0.9063	0.9912	−0.0038	*tied within noise*
v1 + Focal	0.9807	0.9128	0.9907	−0.0026	*tied within noise*
v1 + MixUp	0.9804	0.9096	0.9884	−0.0029	*tied within noise*
v1 + TTA	**0.9854**	**0.9329**	**0.9934**	**+0.0021**	***free improvement***
v2 (all four)	0.9829	0.9212	0.9917	−0.0004	***combined model***

What the ablation tells us

Individually, **CBAM, Focal Loss, and MixUp** are each statistically tied with v1 (Δ within ±0.008 F1 confidence interval).
Only **TTA** delivers a measurable improvement on its own.
But each component addresses a **different limitation** of ResNet-50 — and combining them in v2 produces the ***methodologically complete model.***

Which Model Should We Deploy?

OUR PICK

V2

ResNet-50 + CBAM

Focal Loss · MixUp · TTA

Test metrics

F1 = 0.9829

MCC = 0.9212

ROC-AUC = 0.9917

Why v2 — combining the four components into one model

1. **Methodologically complete**

   v2 addresses every known ResNet-50 limitation in one package: no attention → CBAM, naive imbalance handling → Focal Loss, weak regularisation → MixUp, single-pass inference → TTA.

2. **Interpretability you can show clinicians**

   Only v2 (and v3) produces the CBAM spatial-attention heatmap. v1 + TTA cannot show where the model is looking — a deal-breaker for clinical adoption.

3. **Defence in depth against deployment shift**

   Focal Loss + MixUp + CBAM together regularise the model against new scanners, new populations, or shifted nodule prevalence — v1 + TTA only adds inference-side averaging.

4. **Statistically equivalent on this test set**

   All deltas (v1, v1+TTA, v2) sit within the ±0.008 F1 confidence interval. v2's marginal numerical deficit is not a practical disadvantage — its qualitative gains are.

Alternatives — **v1** for a quick prototype · **v3** for edge / mobile / low-VRAM deployment (3× smaller, similar accuracy).

What we delivered

1

Leakage-free LUNA16 classifier

Patient-level split across full LUNA16 (888 patients). Test = 1 528 patches across 176 unseen patients.

2

Three honestly-compared models

v1 (ResNet-50) · v2 (+CBAM, Focal, MixUp, TTA) · v3 (DenseNet-121+CBAM, Focal, MixUp, TTA)

3

Production-grade results

F1 = 0.9833 · ROC-AUC = 0.9929 · MCC = 0.9233 — competitive with peer-reviewed LUNA16 baselines.

4

Honest scientific story

Detected and corrected patient-level leakage on IQ-OTH/NCCD. Reported deltas with confidence intervals — not just point estimates.

Live Demo (≤ 1 minute)

[Demo video — to be inserted]

Suggested content: load best model → infer on a held-out test patch → show predicted probability and CBAM heatmap overlay (max 60 s).

Team Roles & What Each Member Did

Member	Student ID	Role	Contribution
Huu Thuc Tran	26164741	***Leader***	Dataset acquisition + extraction; patient-leakage analysis on IQ-OTH/NCCD; LUNA16 patch pipeline; implementation, training and evaluation of v1, v2, v3 (CBAM, Focal Loss, MixUp, TTA); 9-metric comparison study; final report.
Alex	26131779	*Member*	Slide design (Seminar 1) and initial Seminar 2 draft; early literature review on lung cancer detection.
Rabiul	*[Student ID — TBC]*	*Member*	Literature review on lung-nodule classification methods.

Team-roles statement: the Leader handled implementation and analytical work; teammates contributed to communication and literature support. *All members agreed the contribution split above is accurate.*

Thank you

Questions & Discussion